EXTENSION 1

Stage 6

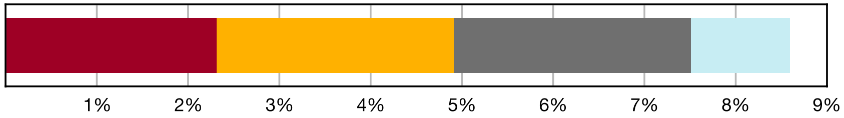
Functions (Ext1), F1 Further Work With Functions (Y11)

Inverse Functions (Ext1)

Teacher: SHS Maths

Exam Equivalent Time: 112.5 minutes *(based on allocation of 1.5 minutes per mark)*

F1 Further Work With Functions



Graphical Relationships
Inequalities
Inverse Functions
Parametric Functions

***SmarterMaths analytics based on the average contribution to Extension 1 exams since 2020.**

HISTORICAL CONTRIBUTION

- *F1 Further Work With Functions* has contributed a healthy 8.6% per new syllabus Ext1 exam since it was introduced in 2020.
- This topic has been split into four categories for analysis purposes which are: *1-Graphical Relationships (2.3%), 2-Inequalities (2.6%), 3-Inverse Functions (2.6%) and 4-Parametric Functions (1.1%).*
- This analysis looks at *Inverse Functions.*

HSC ANALYSIS - What to expect and common pitfalls

- *Inverse Functions (2.6%)* looks at non-trigonometric inverse functions and has been examined in 3 new syllabus exams to date, most recently in 2023.
- When *Inverse Functions* are tested, significant mark allocations are common. *2021 Ext1 12d* is case in point, contributing 6 marks to the exam – a "must review" question.
- Multiple choice questions on this topic have reached band 5-6 difficulty (see *2023 Ext1 9 MC* and *2019 Ext1 10 MC).*
- We note in past years, allocations between this topic and the larger *Inverse Trig Functions* topic tend to cannibalise each other.
- *Pitfalls:* The graph and associated range and domain of an inverse exponential function has been the most poorly answered question type in this area and harder examples deserve revision attention (see *2018 Ext1 13b* and *2010 Ext1 3b).*

Questions

1. Functions, EXT1 F1 2020 HSC 2 MC

Given $f(x) = 1 + \sqrt{x}$, what are the domain and range of $f^{-1}(x)$

A. $x \geq 0, \quad y \geq 0$

B. $x \geq 0, \quad y \geq 1$

C. $x \geq 1, \quad y \geq 0$

D. $x \geq 1, \quad y \geq 1$

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2. Functions, EXT1 F1 2019 MET2-N 11 MC

The function $f(x) = 5x^3 + 10x^2 + 1$ will have an inverse function for the domain

A. $D = (-2, \infty)$

B. $D = \left(-\infty, \dfrac{1}{2}\right]$

C. $D = (-\infty, -1]$

D. $D = [0, \infty)$

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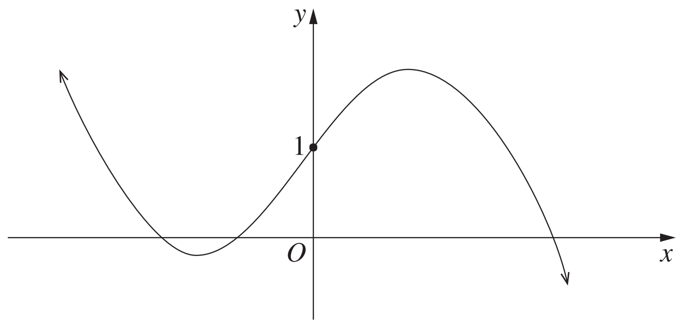
3. Functions, EXT1 F1 2023 HSC 9 MC

The graph of a cubic function, $y = f(x)$, is given below.



Which of the following functions has an inverse relation whose graph has more than 3 points with an x-coordinate of 1 ?

A. $y = \sqrt{f(x)}$

B. $y = \dfrac{1}{f(x)}$

C. $y = f(|x|)$

D. $y = |f(x)|$

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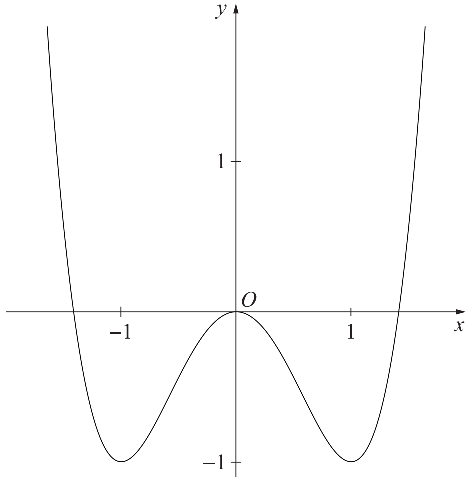
4. Functions, EXT1 F1 2019 HSC 10 MC

The function $f(x) = -\sqrt{1+\sqrt{1+x}}$ has inverse $f^{-1}(x)$.

The graph of $y = f^{-1}(x)$ forms part of the curve $y = x^4 - 2x^2$.

The diagram shows the curve $y = x^4 - 2x^2$.



How many points do the graphs of $y = f(x)$ and $y = f^{-1}(x)$ have in common?

A. 1

B. 2

C. 3

D. 4

5. Functions, EXT1 F1 2016 HSC 11a

Find the inverse of the function $y = x^3 - 2$. ***(2 marks)***

6. Functions, EXT1 F1 2019 MET1-N 5

Let $h(x) = \dfrac{7}{x+2} - 3$ for $x \geq 0$.

a. State the range of h. ***(1 mark)***

b. Find the rule for h^{-1}. ***(2 marks)***

7. Functions, EXT1 F1 2012 HSC 12b

Let $f(x) = \sqrt{4x-3}$

i. Find the domain of $f(x)$. ***(1 mark)***

ii. Find an expression for the inverse function $f^{-1}(x)$. ***(2 marks)***

iii. Find the points where the graphs $y = f(x)$ and $y = x$ intersect. ***(1 mark)***

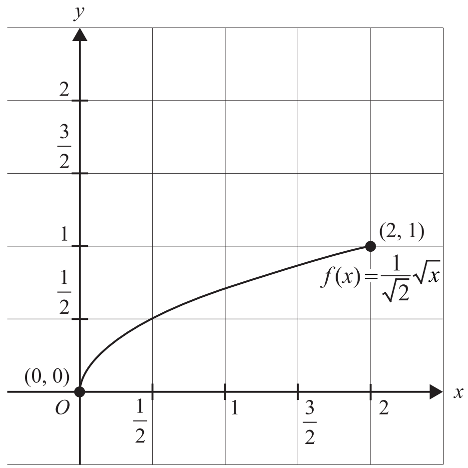
iv. On the same set of axes, sketch the graphs $y = f(x)$ and $y = f^{-1}(x)$ showing the information found in part (iii). ***(2 marks)***

8. Functions, EXT1 F1 2020 MET1 6

$f(x) = \dfrac{1}{\sqrt{2}}\sqrt{x}$, where $x \in [0, 2]$

a. Find $f^{-1}(x)$, and state its domain. ***(2 marks)***

The graph of $y = f(x)$, where $x \in [0, 2]$, is shown on the axes below.



b. On the axes above, sketch the graph of $f^{-1}(x)$ over its domain. Label the endpoints and point(s) of intersection with $f(x)$, giving their coordinates. ***(2 marks)***

9. Functions, EXT1 F1 2019 MET1 5b

Given the function $h(x) = \sqrt{2x+3} - 2$ for $h \geq -\dfrac{3}{2}$, find the inverse function h^{-1} and its domain. ***(3 marks)***

10. Functions, EXT1 F1 2021 HSC 12d

A function is defined by $f(x) = 4 - \left(1 - \frac{x}{2}\right)^2$ for x in the domain $(-\infty, 2]$.

i. Sketch the graph of $y = f(x)$ showing the x- and y-intercepts. ***(2 marks)***

ii. Find the equation of the inverse function, $f^{-1}(x)$, and state its domain. ***(3 marks)***

iii. Sketch the graph of $y = f^{-1}(x)$. ***(1 mark)***

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11. Functions, EXT1 F1 EQ-Bank 3

If $f(x) = \dfrac{4 - e^{5x}}{3}$, find the inverse function $f^{-1}(x)$. ***(2 marks)***

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12. Functions, EXT1 F1 2008 HSC 5a

Let $f(x) = x - \frac{1}{2}x^2$ for $x \le 1$. This function has an inverse, $f^{-1}(x)$.

i. Sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$ on the same set of axes. (Use the same scale on both axes.) ***(2 marks)***

ii. Find an expression for $f^{-1}(x)$. ***(3 marks)***

iii. Evaluate $f^{-1}\left(\frac{3}{8}\right)$. ***(1 mark)***

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13. Functions, EXT1 F1 2006 HSC 5b

Let $f(x) = \log_e(1 + e^x)$ for all x.

Show that $f(x)$ has an inverse. ***(2 marks)***

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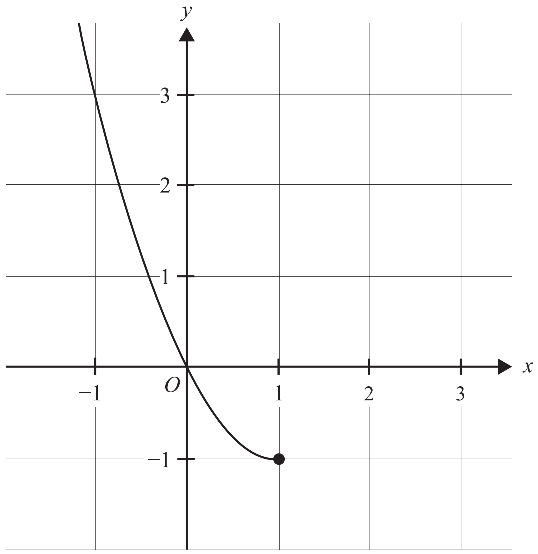
14. Functions, EXT1 F1 2023 MET1 7

Consider $f: (-\infty, 1] \to R,\ f(x) = x^2 - 2x$. Part of the graph of $y = f(x)$ is shown below.



a. State the range of f. ***(1 mark)***

b. Sketch the graph of the inverse function $y = f^{-1}(x)$ on the axes above. Label any endpoints and axial intercepts with their coordinates. ***(2 marks)***

c. Determine the equation of the domain for the inverse function f^{-1}. ***(2 marks)***

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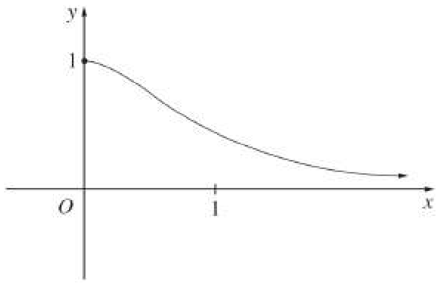
15. Functions, EXT1 F1 2004 HSC 5b*

The diagram below shows a sketch of the graph of $y = f(x)$, where $f(x) = \dfrac{1}{1+x^2}$ for $x \geq 0$.



i. On the same set of axes, sketch the graph of the inverse function, $y = f^{-1}(x)$. ***(1 mark)***

ii. State the domain of $f^{-1}(x)$. ***(1 mark)***

iii. Find an expression for $y = f^{-1}(x)$ in terms of x. ***(2 marks)***

iv. The graphs of $y = f(x)$ and $y = f^{-1}(x)$ meet at exactly one point P.
Let α be the x-coordinate of P. Explain why α is a root of the equation $x^3 + x - 1 = 0$. ***(1 mark)***

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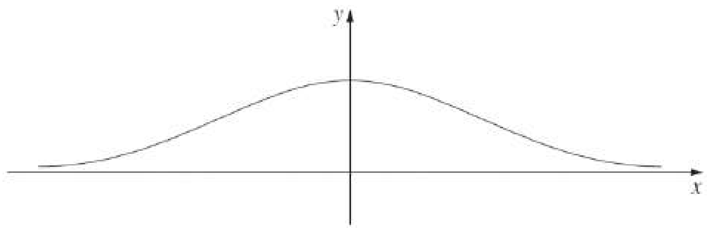
16. Functions, EXT1 F1 2010 HSC 3b*

Let $f(x) = e^{-x^2}$. The diagram shows the graph $y = f(x)$.



i. The graph has two points of inflection.
Find the x coordinates of these points. ***(3 marks)***

ii. Explain why the domain of $f(x)$ must be restricted if $f(x)$ is to have an inverse function. ***(1 mark)***

iii. Find a formula for $f^{-1}(x)$ if the domain of $f(x)$ is restricted to $x \geq 0$. ***(2 marks)***

iv. State the domain of $f^{-1}(x)$. ***(1 mark)***

v. Sketch the curve $y = f^{-1}(x)$. ***(1 mark)***

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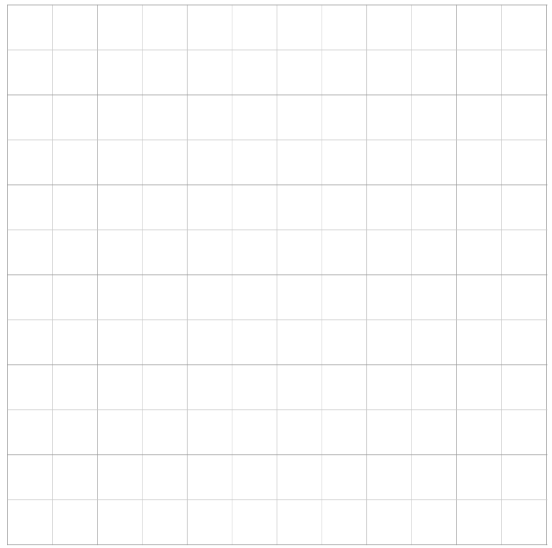
17. Functions, EXT1 F1 EQ-Bank 4

Let $f(x) = x^2 - 4x + 3$ for $x \leqslant 2$.

a. State the domain of $y = f^{-1}(x)$. ***(1 mark)***

b. What is the equation of $y = f^{-1}(x)$? ***(3 marks)***

c. For the restricted domain, sketch the function and the inverse function on the same number plane below.
Clearly identify all axis intercepts. ***(2 marks)***



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18. Functions, EXT1 F1 2009 HSC 3a

Let $f(x) = \dfrac{3+e^{2x}}{4}$.

i. Find the range of $f(x)$. ***(1 mark)***

ii. Find the inverse function $f^{-1}(x)$. ***(2 marks)***

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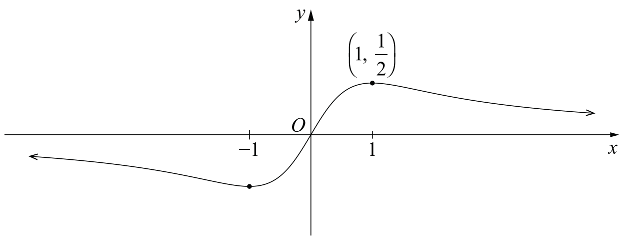
19. Functions, EXT1 F1 2018 HSC 13b

The diagram shows the graph $y = \dfrac{x}{x^2+1}$, for all real x.



Consider the function $f(x) = \dfrac{x}{x^2+1}$, for $x \geq 1$.

The function $f(x)$ has an inverse. (Do NOT prove this.)

i. State the domain and range of $f^{-1}(x)$. ***(2 marks)***

ii. Sketch the graph $y = f^{-1}(x)$. ***(1 mark)***

iii. Find an expression for $f^{-1}(x)$. ***(3 marks)***

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20. Functions, EXT1 F1 2007 HSC 6b

Consider the function $f(x) = e^x - e^{-x}$.

i. Show that $f(x)$ is increasing for all values of x. ***(1 mark)***

ii. Show that the inverse function is given by

$$f^{-1}(x) = \log_e\left(\frac{x+\sqrt{x^2+4}}{2}\right)$$ ***(3 marks)***

iii. Hence, or otherwise, solve $e^x - e^{-x} = 5$. Give your answer correct to two decimal places. ***(1 mark)***

Worked Solutions

1. Functions, EXT1 F1 2020 HSC 2 MC

$\text{Domain}\ \ f(x):\ x \geq 0$

$\text{Range}\ \ f(x):\ y \geq 1$

$\text{Domain}\ \ f^{-1} = \text{Range}\ \ f(x) = x \geq 1$

$\text{Range}\ \ f^{-1} = \text{Range}\ \ f(x) = y \geq 0$

$\Rightarrow C$

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2. Functions, EXT1 F1 2019 MET2-N 11 MC

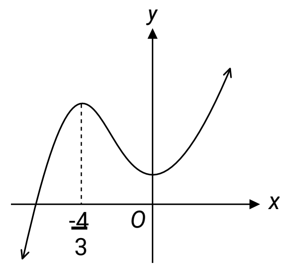
$y = 5x^3 + 10x^2 + 1$

$y' = 15x^2 + 20x$

$y'' = 30x + 20$

$\text{SP's when}\ \ y' = 0 \ \ \Rightarrow \ \ x = -\dfrac{4}{3}\ (\text{max}),\ 0\ (\text{min})$

Sketch graph:



$\Rightarrow\ D$

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3. Functions, EXT1 F1 2023 HSC 9 MC

An inverse function with more than 3 points when $x = 1$ is the same as the original function intersecting $y = 1$ more than 3 times.

♦ Mean mark 47%.

Consider $y = |\,f(x)\,|$:

$y = 1$ cuts $y = f(x)$ three times and $y = |\,f(x)|$ four times as the graph below the x-axis (bottom right) is reflected in the axis.

$\Rightarrow D$

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4. Functions, EXT1 F1 2019 HSC 10 MC

$f(x) = -\sqrt{1 + \sqrt{1+x}}$

Domain $f(x)$ is $x \geq -1$

Range $f(x)$ is $y \leq -1$ since y decreases as x increases.

$f(-1) = -\sqrt{1 + \sqrt{0}} = -1$

$\therefore$ One intersection (only) occurs at $(-1, -1)$.

$\Rightarrow A$

♦♦♦ *Mean mark 29%.*

5. Functions, EXT1 F1 2016 HSC 11a

$y = x^3 - 2$

For inverse, $x \leftrightarrow y$

$x = y^3 - 2$

$y^3 = x + 2$

$\therefore y = (x+2)^{\frac{1}{3}}$

6. Functions, EXT1 F1 2019 MET1-N 5

a. y_{max} occurs when $x = 0$

$y_{\text{max}} = \dfrac{7}{2} - 3 = \dfrac{1}{2}$

As $x \to \infty$, $\dfrac{7}{x+2} \to 0^+$

$\therefore$ Range $h(x) = \left(-3, \dfrac{1}{2}\right)$

b. $y = \dfrac{7}{x+2}$

Inverse: swap $x \leftrightarrow y$

$$x = \frac{7}{y+2} - 3$$

$$x + 3 = \frac{7}{y+2}$$

$$y + 2 = \frac{7}{x+3}$$

$$y = \frac{7}{x+3} - 2$$

$$\therefore h^{-1} = \frac{7}{x+3} - 2$$

7. Functions, EXT1 F1 2012 HSC 12b

i. $f(x) = \sqrt{4x - 3}$

Domain exists when $4x - 3 \geq 0$

$$4x \geq 3$$

$$x \geq \frac{3}{4}$$

ii. Inverse function when

$$x = \sqrt{4y - 3}$$

$$x^2 = 4y - 3$$

$$4y = x^2 + 3$$

$$y = \frac{1}{4}x^2 + \frac{3}{4}$$

Since range of $f(x)$ is $y \geq 0$

$\Rightarrow$ Domain of $f^{-1}(x)$ is $x \geq 0$

$\therefore f^{-1}(x) = \dfrac{1}{4}x^2 + \dfrac{3}{4}, \quad x \geq 0$

iii. Find intersection

$y = \sqrt{4x - 3}$ ... (1)

$y = x$ ... (2)

Intersection occurs when

$$x = \sqrt{4x - 3}$$

$$x^2 = 4x - 3$$

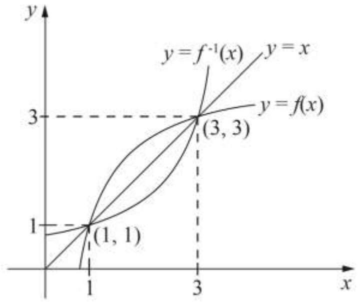
$$x^2 - 4x + 3 = 0$$

$$(x-3)(x-1) = 0$$

$$x = 1 \text{ or } 3$$

$\therefore$ Intersection at $(1, 1)$ and $(3, 3)$

iv.



8. Functions, EXT1 F1 2020 MET1 6

a. Domain $f^{-1}(x)$ = Range $f(x) = [0, 1]$

$$y = \frac{1}{\sqrt{2}}x$$

Inverse: swap $x \leftrightarrow y$

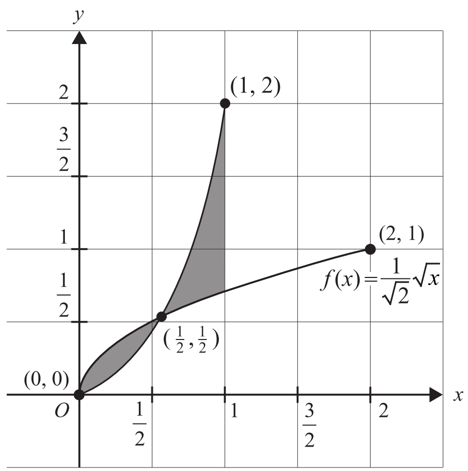
$$x = \frac{1}{\sqrt{2}}\sqrt{y}$$

$$\sqrt{y} = \sqrt{2}x$$

$$y = 2x^2$$

$$\therefore f^{-1}(x) = 2x^2$$

b.



9. Functions, EXT1 F1 2019 MET1 5b

$$y = \sqrt{2x+3} - 2$$

Inverse: swap $x \leftrightarrow y$

$$x = \sqrt{2y+3} - 2$$

$$\sqrt{2y+3} = x + 2$$

$$2y + 3 = (x+2)^2$$

$$y = \frac{1}{2}(x+2)^2 - \frac{3}{2}$$

$$\therefore h^{-1} = \frac{1}{2}(x+2)^2 - \frac{3}{2}$$

Domain $h^{-1}(x)$ = Range $h(x)$

$= x \geq -2$ or $[-2, \infty)$

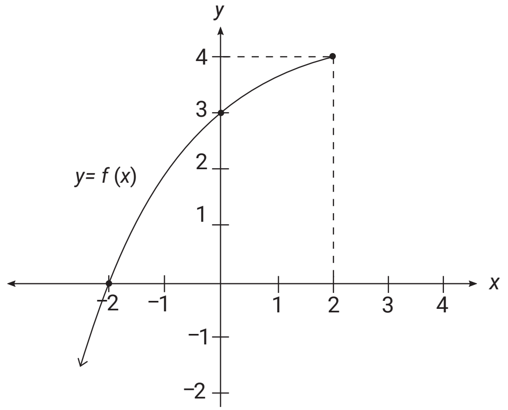
10. Functions, EXT1 F1 2021 HSC 12d

i. $f(x) = 4 - \left(1 - \dfrac{x}{2}\right)^2$

$\text{At } \ x = 2, \ f(x) = 4$

$\text{At } \ x = 0, \ f(x) = 3$

$x\text{-intercept: } \ 1 - \dfrac{x}{2} = 2 \ \rightarrow \ x = -2$



ii. Inverse function: swap $x \leftrightarrow y$

$$x = 4 - \left(1 - \frac{y}{2}\right)^2$$

$$\left(1 - \frac{y}{2}\right)^2 = 4 - x$$

$$1 - \frac{y}{2} = \pm\sqrt{4 - x}$$

$$\frac{y}{2} = 1 \pm \sqrt{4 - x}$$

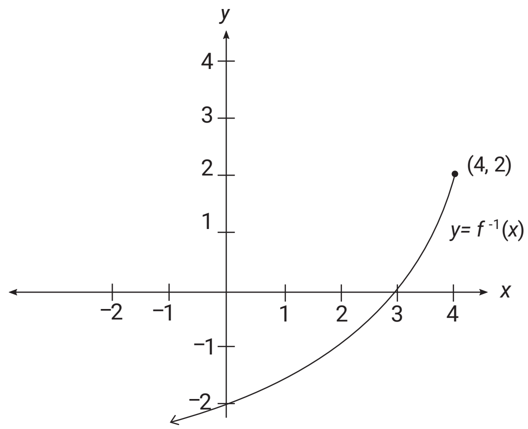
$$y = 2 \pm 2\sqrt{4 - x}$$

$$\therefore f^{-1}(x) = 2 - 2\sqrt{4 - x}, \quad (y \le 2)$$

$$\text{Domain } \ f^{-1}(x) = \text{Range } f(x)$$

$$= x \in (-\infty, 4]$$

iii.



11. Functions, EXT1 F1 EQ-Bank 3

$$f(x) = \frac{4 - e^{5x}}{3}$$

Inverse: swap $x \leftrightarrow y$

$$x = \frac{4 - e^{5x}}{3}$$

$$3x = 4 - e^{5y}$$

$$e^{5y} = 4 - 3x$$

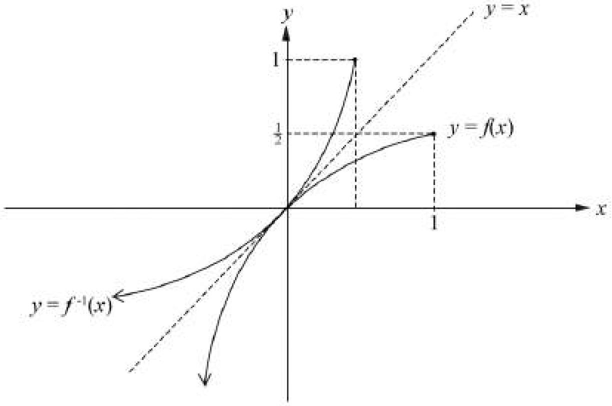
$$\ln e^{5y} = \ln|4 - 3x|$$

$$5y = \ln|4 - 3x|$$

$$y = \frac{1}{5}\ln|4 - 3x|$$

12. Functions, EXT1 F1 2008 HSC 5a

i.



ii. $y = x - \dfrac{1}{2}x^2, \quad x \le 1$

Inverse function: swap $x \leftrightarrow y$,

$$x = y - \frac{1}{2}y^2, \quad y \le 1$$

$$2x = 2y - y^2$$

$$y^2 - 2y + 2x = 0$$

$$\begin{aligned} y &= \frac{2 \pm \sqrt{(-2)^2 - 4 \cdot 1 \cdot 2x}}{2} \\ &= \frac{2 \pm \sqrt{4 - 8x}}{2} \\ &= \frac{2 \pm 2\sqrt{1 - 2x}}{2} \\ &= 1 \pm \sqrt{1 - 2x} \end{aligned}$$

$$\therefore\ y = 1 - \sqrt{1 - 2x}, \quad (y \le 1)$$

iii. $$f^{-1}\left(\frac{3}{8}\right) = 1 - \sqrt{1 - 2\left(\frac{3}{8}\right)}$$

$$\begin{aligned} &= 1 - \sqrt{1 - \frac{6}{8}} \\ &= 1 - \sqrt{\frac{1}{4}} \\ &= 1 - \frac{1}{2} \\ &= \frac{1}{2} \end{aligned}$$

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13. Functions, EXT1 F1 2006 HSC 5b

$$f(x) = \log_e(1 + e^x)$$

$$f\prime(x) = \frac{e^x}{1 + e^x}$$

$\Rightarrow e^x > 0$ for all x

$\therefore\ \dfrac{e^x}{1 + e^x} > 0$ for all x

$\therefore\ f(x)$ is a monotonic increasing function for all x.

$\therefore\ f(x)$ has an inverse for all x.

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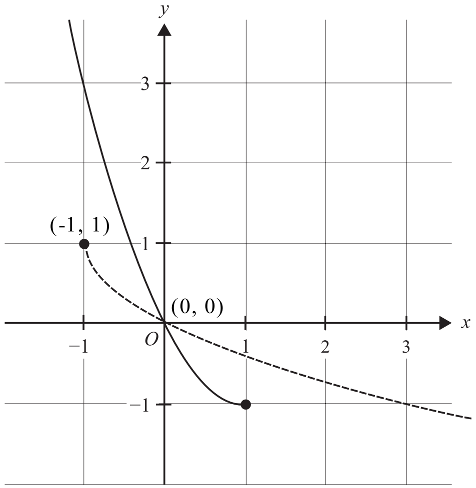
14. Functions, EXT1 F1 2023 MET1 7

a. $[-1, \infty)$

b.



c. When $f(x)$ is written in turning point form

$y = (x-1)^2 - 1$

Inverse function: swap $x \leftrightarrow y$

$$x = (y-1)^2 - 1$$
$$x + 1 = (y-1)^2$$
$$-\sqrt{x+1} = y - 1$$
$$f^{-1}(x) = 1 - \sqrt{x+1}$$

Domain $[-1, \infty)$

♦ Mean mark (c) 48%.
MARKER'S COMMENT:
Common error → writing the function as $f^{-1}(x) = 1 + \sqrt{x+1}$.

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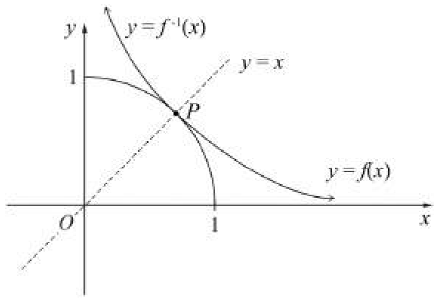
15. Functions, EXT1 F1 2004 HSC 5b*

i.



ii. Domain of $f^{-1}(x)$ is

$0 < x \leq 1$

iii. $f(x) = \dfrac{1}{1+x^2}$

Inverse: swap $x \leftrightarrow y$

$$x = \frac{1}{1+y^2}$$
$$x\left(1+y^2\right) = 1$$
$$1 + y^2 = \frac{1}{x}$$
$$y^2 = \frac{1}{x} - 1$$
$$= \frac{1-x}{x}$$
$$y = \pm\sqrt{\frac{1-x}{x}}$$

$$\therefore y = \sqrt{\frac{1-x}{x}}, \quad y \geq 0$$

iv. P occurs when $f(x)$ cuts $y = x$

i.e. where

$$\frac{1}{1+x^2} = x$$
$$1 = x\left(1+x^2\right)$$
$$1 = x + x^3$$
$$x^3 + x - 1 = 0$$

$\Rightarrow$ Since α is the x-coordinate of P, it is a root of $x^3 + x - 1 = 0$

16. Functions, EXT1 F1 2010 HSC 3b*

i.

$$y = e^{-x^2}$$

$$\frac{dy}{dx} = -2x \cdot e^{-x^2}$$

$$\frac{d^2y}{dx^2} = -2x\left(-2x \cdot e^{-x^2}\right) + e^{-x^2}(-2)$$

$$= 4x^2e^{-x^2} - 2e^{-x^2}$$

$$= 2e^{-x^2}\left(2x^2 - 1\right)$$

P.I. when $\dfrac{d^2y}{dx^2} = 0$

$$2e^{-x^2}\left(2x^2 - 1\right) = 0$$

$$2x^2 - 1 = 0$$

$$x^2 = \frac{1}{2}$$

$$x = \pm\frac{1}{\sqrt{2}}$$

When $x < \dfrac{1}{\sqrt{2}}, \ \dfrac{d^2y}{dx^2} < 0$

$x > \dfrac{1}{\sqrt{2}}, \ \dfrac{d^2y}{dx^2} > 0$

$\Rightarrow$ Change of concavity

$\therefore$ P.I. at $x = \dfrac{1}{\sqrt{2}}$

When $x < -\dfrac{1}{\sqrt{2}}, \ \dfrac{d^2y}{dx^2} > 0$

$x > -\dfrac{1}{\sqrt{2}}, \ \dfrac{d^2y}{dx^2} < 0$

$\Rightarrow$ Change of concavity

$\therefore$ P.I. at $x = -\dfrac{1}{\sqrt{2}}$

COMMENT: It is also valid to show that $f(x)$ is an even function and if a P.I. exists at $x = a$, ***there must*** be another P.I. at $x = -a$.

ii. In $f(x)$, there are 2 values of y for each value of x.

$\therefore$ The domain of $f(x)$ must be restricted for $f^{-1}(x)$ to exist.

iii. $y = e^{-x^2}$

Inverse function can be written

x

$$= e^{-y^2}, \quad x \geq 0$$

$$\ln x = \ln e^{-y^2}$$

$$-y^2 = \ln x$$

$$y^2 = -\ln x$$

$$= \ln\left(\frac{1}{x}\right)$$

$$y = \pm\sqrt{\ln\left(\frac{1}{x}\right)}$$

Restricting $x \geq 0, \quad \Rightarrow y \geq 0$

$$\therefore\ f^{-1}(x) = \sqrt{\ln\left(\frac{1}{x}\right)}$$

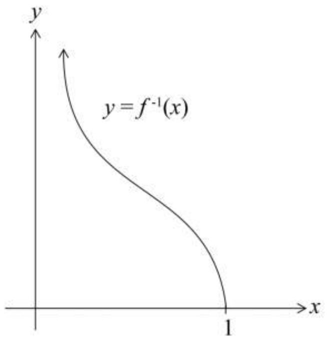
iv. $f(0) = e^0 = 1$

$\therefore$ Range of $f(x)$ is $0 < y \leq 1$

$\therefore$ Domain of $f^{-1}(x)$ is $0 < x \leq 1$

♦ *Parts (iv) and (v) were poorly answered with mean marks of 39% and 49% respectively.*

v.



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17. Functions, EXT1 F1 EQ-Bank 4

a. $f(x) = (x-1)(x-3) \ \Rightarrow$ Axis of symmetry at $x = 2$.

$f(2) = -1$

Range $f(x):\ y \geqslant -1 \ \Rightarrow$ Domain $f^{-1}(x):\ x \geqslant -1$

b. Inverse: swap $x \leftrightarrow y$

$$x = y^2 - 4y + 3$$

$$x = (y-2)^2 - 1$$

$$x + 1 = (y-2)^2$$

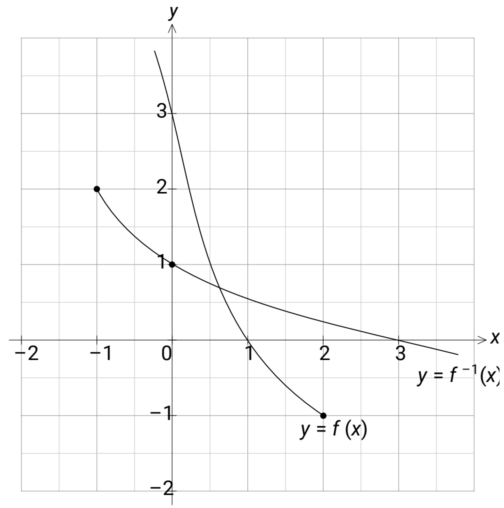
$$y - 2 = \pm\sqrt{x+1}$$

$$y = 2 \pm \sqrt{x+1}$$

$$y = 2 - \sqrt{x+1} \quad (\text{Range } f^{-1}(x):\ y \leqslant 2)$$

c. $f(x)$ intercepts: $(1, 0), (0, 3)$

$f^{-1}(x)$ intercepts: $(3, 0), (0, 1)$



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18. Functions, EXT1 F1 2009 HSC 3a

i. $f(x) = \dfrac{3+e^{2x}}{4}$

As $x \rightarrow \infty,\ e^{2x} \rightarrow \infty,\ f(x) \rightarrow \infty$

As $x \rightarrow -\infty,\ e^{2x} \rightarrow 0,\ f(x) \rightarrow \dfrac{3}{4}$

$\therefore$ Range is $y > \dfrac{3}{4}$

ii. Inverse function: swap $x \leftrightarrow y$

$$\begin{aligned} x &= \frac{3+e^{2y}}{4} \\ 4x &= 3+e^{2y} \\ e^{2y} &= 4x-3 \\ \ln e^{2y} &= \ln(4x-3) \\ 2y &= \ln(4x-3) \\ y &= \frac{1}{2}\ln(4x-3) \end{aligned}$$

$\therefore\ f^{-1}(x) = \dfrac{1}{2}\ln(4x-3)$

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19. Functions, EXT1 F1 2018 HSC 13b

i. Domain of $f(x)$ is $x \geq 1$ (given)

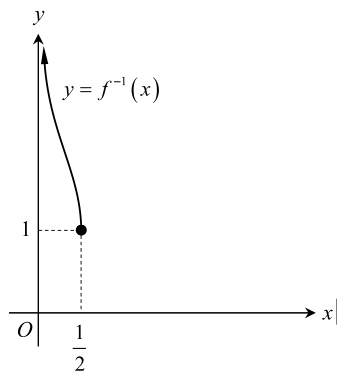
$\Rightarrow$ Range of $f^{-1}(x)$ is $y \geq 1$

♦♦ *Mean mark part (i) 31%.*

Range of $f(x)$ is $0 < y < \dfrac{1}{2}$

$\Rightarrow$ Domain of $f^{-1}(x)$ is $0 < x \leq \dfrac{1}{2}$.

ii.



♦ *Mean mark part (ii) 43%.*

iii. $y = \dfrac{x}{x^2+1}$

Inverse: $x \leftrightarrow y$

♦ *Mean mark part (iii) 45%.*

$$x = \frac{y}{y^2+1}$$

$$xy^2 + x = y$$

$$xy^2 - y + x = 0$$

$$y = \frac{1 \pm \sqrt{1-4x^2}}{2x}$$

Since, range of $f^{-1}(x)$ is $y \geq 1$,

$\therefore\ f^{-1}(x) = \dfrac{1+\sqrt{1-4x^2}}{2x}$

20. Functions, EXT1 F1 2007 HSC 6b

i. $f(x) = e^x - e^{-x}$

$f\prime(x) = e^x + e^{-x}$

$$\text{Since} \quad e^x > 0 \ \text{ for all } x$$
$$e^{-x} > 0 \ \text{ for all } x$$
$$f\prime(x) > 0 \ \text{ for all } x$$

$\therefore f(x)$ is an increasing function for all x.

ii. $y = e^x - e^{-x}$

Inverse function

$$x = e^y - \frac{1}{e^y}$$
$$xe^y = e^{2y} - 1$$
$$e^{2y} - xe^y - 1 = 0$$

Let $A = e^y$

$\therefore A^2 - xA - 1 = 0$

Using the quadratic formula

$$A = \frac{x \pm \sqrt{(-x)^2 - 4 \cdot 1 \cdot (-1)}}{2 \cdot 1}$$
$$= \frac{x \pm \sqrt{x^2 + 4}}{2}$$

Since $\dfrac{x - \sqrt{x^2+4}}{2} < 0$ and $e^y > 0$,

$$\therefore e^y = \frac{x + \sqrt{x^2+4}}{2}$$
$$\log_e e^y = \log_e\left(\frac{x + \sqrt{x^2+4}}{2}\right)$$
$$y = \log_e\left(\frac{x + \sqrt{x^2+4}}{2}\right)$$
$$\therefore f^{-1}(x) = \log_e\left(\frac{x + \sqrt{x^2+4}}{2}\right) \quad \ldots \text{ as required}$$

iii. $e^x - e^{-x} = 5$

$= 5$

$f(x)$

$f^{-1}(5) = x$

$$f^{-1}(5) = \log_e\left(\frac{5+\sqrt{5^2+4}}{2}\right)$$

$$= \log_e\left(\frac{5+\sqrt{29}}{2}\right)$$

$$= 1.647\ldots$$

$$= 1.65 \quad \text{(to 2 d.p.)}$$